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İSTANBUL KÜLTÜR UNIVERSITY

FACULTY OF ENGINEERING

GRADUATING PROJECT EVALUATION APPLICATION

SOFTWARE DEVELOPMENT WITH STANDARDS

Sprint 1 – WP-01

Mobile Interface & Clickable Prototype

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1 – Introduction

The aim of this task is to design and implement a mobile user interface that enables jury members to log into the system, view assigned projects, select students, and enter evaluation scores based on predefined criteria. In addition, a clickable prototype was developed to demonstrate the interaction flow between screens.

The main objective of this work is to create a clear and usable evaluation process on mobile devices by structuring the scoring workflow and ensuring that jury members can complete their evaluation tasks efficiently.

2 – System Design

The overall system is designed using a layered architecture that separates the mobile user interface, backend services, and database components. This approach helps to clearly define system responsibilities and supports modular development.

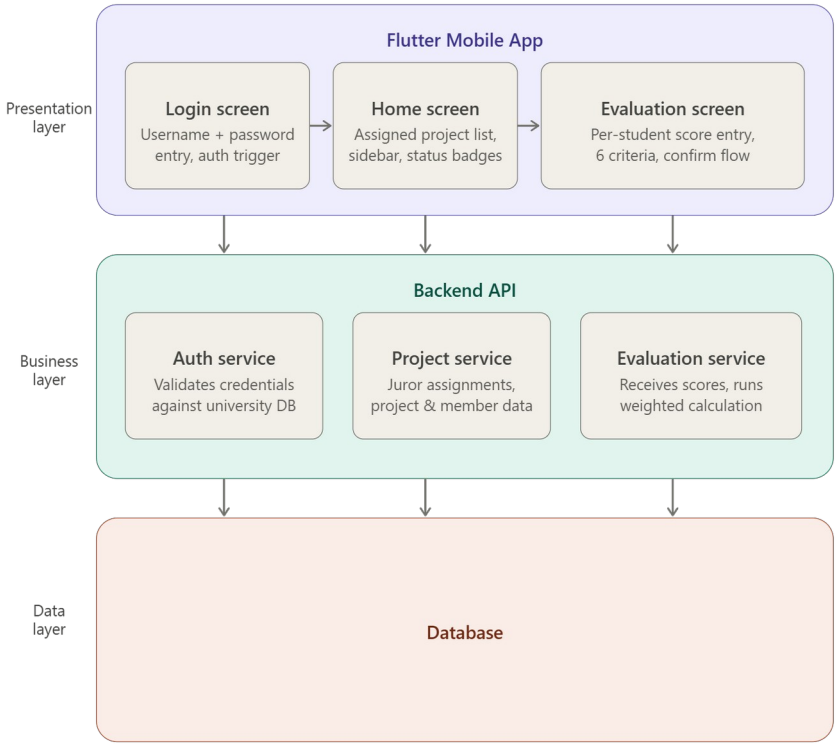


Figure 1 System architecture diagram

Figure 1 shows the general system architecture of the proposed capstone evaluation application. The architecture consists of three main layers: presentation layer, business layer, and data layer.

At the presentation layer, a Flutter-based mobile application provides the main interaction point for jury members. The application includes three primary screens: login, home (assigned project list), and evaluation. Through these screens, users can authenticate, review their assigned projects, and enter evaluation scores for individual students.

The business layer is implemented through backend API services that manage core system operations. The authentication service verifies user credentials by communicating with the university system. The project service handles juror–project assignments and provides project and student information to the mobile interface. The evaluation service processes submitted scores and performs the necessary calculations for final evaluation results.

The data layer consists of a relational database that stores entities such as users, projects, students, juror assignments, and evaluation records. This structure ensures that evaluation data is stored consistently and that jury members can only access information related to their assigned responsibilities.

This layered system design enables clear communication between components and supports a structured evaluation workflow within the mobile application.

3 – User Flow Design

The user flow diagram represents the step-by-step interaction process that a jury member follows while using the mobile evaluation application. This flow focuses on how users navigate between screens and complete the grading task.

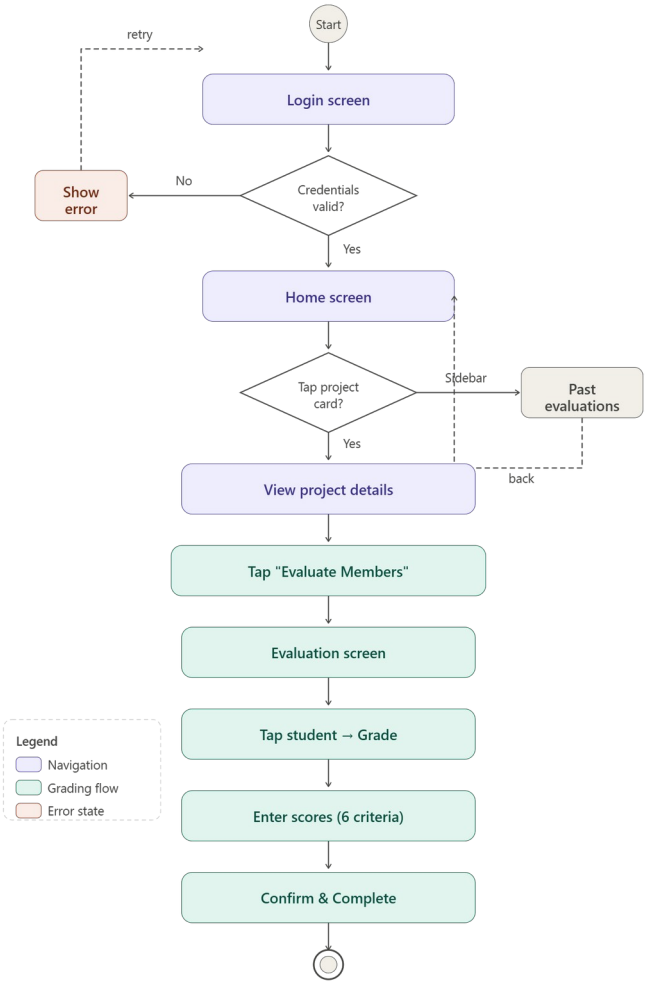


Figure 2 User flow diagram

Figure 2 shows the overall user flow of the system. The process starts with the login screen, where the jury member enters credentials. If the credentials are invalid, an error message is displayed and the user is prompted to retry. After successful authentication, the user is directed to the home screen displaying assigned projects.

From the home screen, the jury member selects a project card to view detailed project information. At this stage, users can either proceed to the evaluation process or access additional options such as viewing past evaluations through the sidebar navigation.

When the user chooses to evaluate a project, the system navigates to the evaluation screen. The jury member then selects a student and enters scores based on predefined evaluation criteria. After all required scores are provided, the evaluation is finalized through a confirmation step.

This user flow ensures that the evaluation process is completed in a clear and sequential manner, reducing confusion and supporting efficient task completion on mobile devices.

4 – Mobile Interface Design & Prototype Implementation

The mobile interface prototype was implemented using the Flutter framework and developed within the Visual Studio Code environment. The objective of this implementation was to create a clickable prototype that demonstrates the intended navigation structure and evaluation workflow of the system.

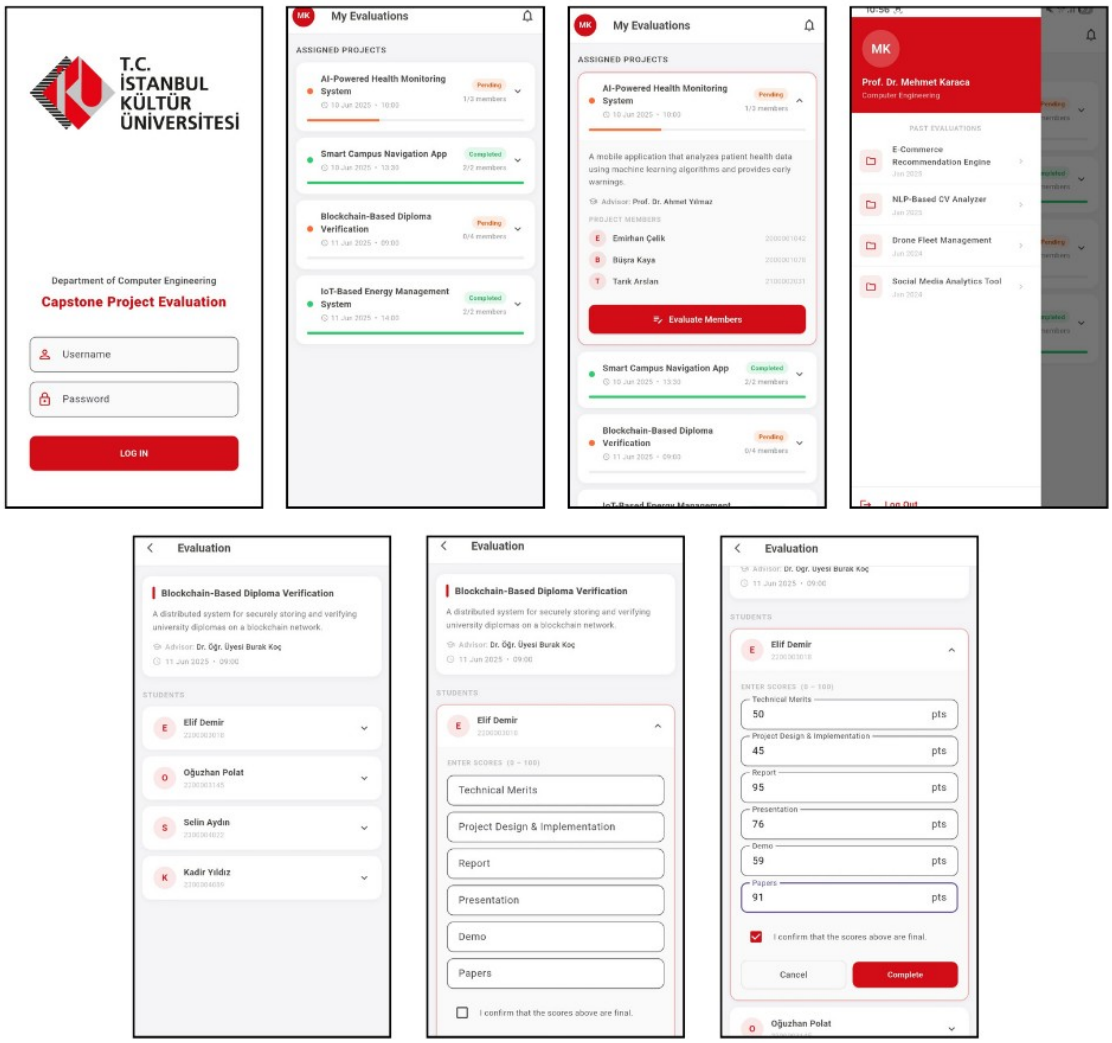


Figure 3 Screenshots of the prototype

Figure 3 presents the main screens of the mobile prototype, including the login interface, assigned project list, project detail page, and student evaluation screens. These screens illustrate how jury members interact with the application during the grading process.

The prototype supports basic screen transitions such as navigating from login to the home screen, selecting a project, and accessing the evaluation interface. However, backend service integration was not implemented in this stage. As a result, operations such as credential validation, automatic score calculation, and dynamic data retrieval are not functionally active.

To make the workflow understandable and visually clear, sample project and student information was included as dummy data. The interface components and navigation structure were developed in a modular way so that backend APIs can be integrated in future development phases without major structural changes.

This prototype demonstrates the usability and feasibility of the proposed mobile evaluation interface and provides a practical representation of the designed user experience.